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USN | | | | | | | | | | [23BSPH011]

FIRST SEMESTER B. Tech EXAMINATION, JANUARY-2025

ENGINEERING PHYSICS

Time: 3 Hours **Maximum Marks: 100**

Instructions:
- **i.** Missing data may be suitably assumed.
- **ii.** Physical constants
  - Mass of the electron $m_e = 9.1 \times 10^{-31} kg$
  - Boltzmann constant $k = 1.38 \times 10^{-23} J/K$
  - Charge of the electron $e = 1.602 \times 10^{-19} C$
  - Planck's constant $h = 6.626 \times 10^{-34} Js$
  - Velocity of light $c = 3 \times 10^{8} m/s$
- **iii.** Answer all questions

		PART - A			**05 X 02=10**
1.	**a**	Explain the concept of wave-particle dualism.	L-2	CO1	**02**
	b.	Why population inversion is essential for laser action?	L-3	CO2	**02**
	c.	Define Fermi Energy.	L-1	CO3	**02**
	d.	What is Simple harmonic motion?	L-1	CO4	**02**
	e.	State Hooke's law in elasticity.	L-1	CO5	02
		PART - B			**06 X 05=30**
2.	**a.**	Derive an expression for de-Broglie wavelength.	L-2	CO1	**05**
		OR			
	b.	Discuss the properties of matter waves and explain Heisenberg's uncertainty principle.	L-2	CO1	**05**
3.	**a.**	Define the wave function and explain the physical significance of wave function.	L-2	CO1	**05**
		OR			
	b.	With neat labelled diagrams, explain the interaction of radiation with matter.	L-3	CO2	**05**
4.	**a.**	Describe the mechanism of light propagation in optical fibers using total internal reflection with the help of a neat diagram.	L-3	CO2	**05**
		OR			
	b.	Discuss any two types of optical fibers with neat diagrams.	L-3	CO2	**05**
5.	**a.**	Mention the assumptions of classical free electron theory of metals.	L-3	CO3	**05**
		OR			

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	b.	Discuss the quantum free electron theory and how it overcomes the drawbacks of classical theory.	L-3	CO3	**05**
6.	**a.**	Derive the differential equation for Simple Harmonic Motion.	L-2	CO5	**05**
		OR			
	b.	Describe the construction and working of an LED with a diagram.	L-3	CO4	**05**
7.	**a.**	Define stress and strain. Paraphrase the types of elastic moduli.	L-2	CO6	**05**
		OR			
	b.	What is a beam? Explain the concept of bending moment.	L-4	CO6	**05**
		PART - C		**06 X 10=60**	
8.	**a.**	**i.** Describe Davisson and Germer`s experiment for confirmation of de-Broglie's hypothesis.	L-2	CO1	**08**
		ii. A bullet of mass 50 g is moving with a speed of 500 m/s. Find its de-Broglie wavelength.	L-3	CO1	**02**
		OR			
	b.	**i.** Prove the non-existence of an electron inside the nucleus using Heisenberg's uncertainty principle.	L-3	CO1	**07**
		ii. In a measurement of position and momentum, the uncertainty in the measurement of position involved is $\Delta x=1.0\times10^{-10}$ m. What is minimum uncertainty in its momentum?	L-3	CO1	**03**
9.	**a.**	Derive Schrödinger's time-independent wave equation in one dimension.	L-3	CO1	**10**
		OR			
	b.	Using one dimensional time independent Schrodinger's wave equation, discuss the solutions for energy of a particle in one dimensional potential well of infinite height.	L-4	CO1	**10**
10.	**a.**	**i.** Describe the working of a point-to-point communication system using optical fibers.	L-3	CO2	**5**
		ii. Explain the two types of attenuation mechanism in an optical fiber.	L-2	CO2	**5**
		OR			
	b.	Describe the construction and working of Nd-YaG Laser with energy level diagram	L-4	CO2	**10**

11.	**a.**	**i.** Distinguish between conductors, semiconductors, and insulators based on band theory.	.L-2	CO4	**06**
		ii. Use the following data for intrinsic silicon at 350 K. Find the resistivity. Intrinsic carrier concentration, $n_i = 1.0\times10^{-16}m^{-3}$, Electron mobility $\mu_e = 0.13\ m^2V^{-1}S^{-1}$, Hole mobility $\mu_h = 0.035m^2V^{-1}S^{-1}$	L-3	CO4	**04**
		OR			
	b.	Discuss the procedure to obtain plank's constant using LED's experimentally with necessary equations.	L-2	CO4	**10**
12.	**a.**	Obtain an expression for force constant in case of springs when connected in series and parallel combination.	L-2	CO5	**10**
		OR			
	b.	Give the theory of damped oscillations and explain different types of damping with necessary conditions.	L-2	CO5	**10**
13.	**a.**	Obtain the relation between Young's modulus, rigidity modulus and Poisson's ratio when the force is applied on one face of a cube.	L-3	CO6	**10**
		OR			
	b.	**i.** Derive the expression for Young's modulus of a material using a single cantilever.	L-3	CO6	**07**
		ii. A steel wire of length 2m and cross-sectional area $1mm^2$ is stretched by a force of 400N.Find the elongation of the wire.	L-3	CO6	**03**